

BIOMIMICRY • AERODYNAMICS

Noise reduced / aerodynamically improved vehicles and aeroplane designs

Nature-inspired ideas for lower drag, lighter structures and more efficient motion.

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What is Biomimicry?

bios = life mimesis = imitation

Learning from nature — then translating those strategies into human design.

- **3.8 Billion Years of R&D**: Evolutionary pressures eliminate inefficient designs; nature optimizes for maximum efficiency with minimal energy input.
- **Form, Process, & System**: Mimicking physical shapes (e.g., bird wings), chemical processes, or ecosystem-level interactions.
- **Sustainability**: Nature operates on closed-loop energy systems, making bio-inspired designs inherently resource-efficient.

What is Aerodynamics?

Aerodynamics studies how moving air behaves and the forces acting on a body moving through it.



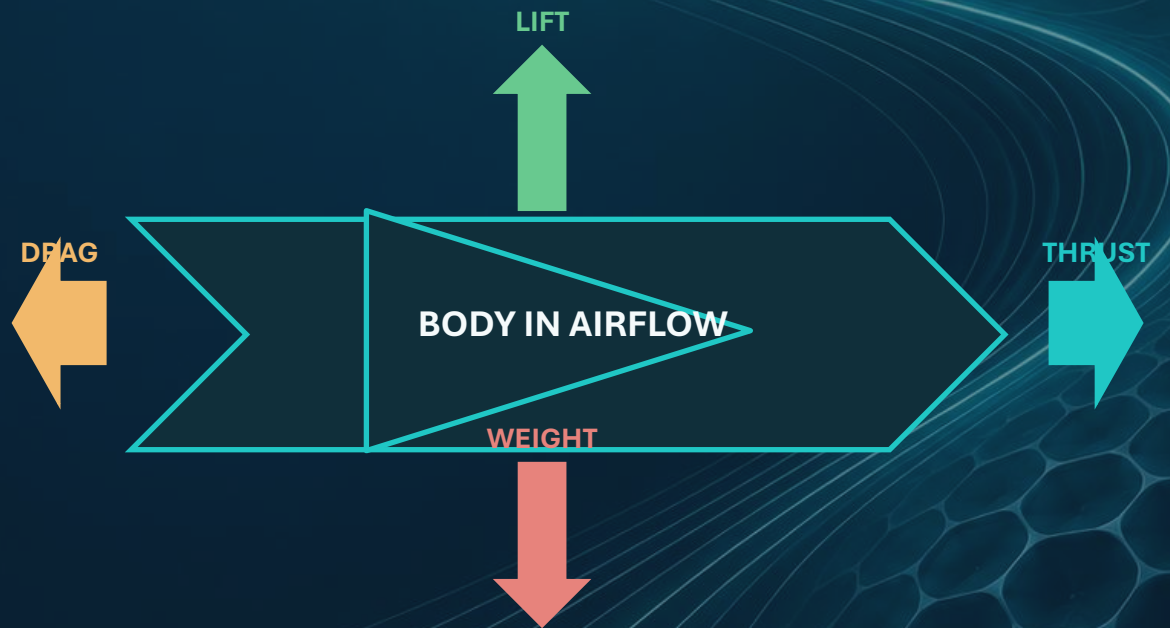
Lift

Upward force from pressure differences and reaction forces.



Drag

Resistance opposing motion: skin-friction, form and induced drag.



Why Biomimicry is Useful in Aerodynamic Design

01

Engineering problem

- Smooth surfaces + rigid geometries
- Boundary-layer turbulence
- High energy dissipation

02

Nature's edge

- Textured surfaces
- Flexible structures
- Passive vortex control
- Works across flow regimes

03

Impact

2–5%

fuel-efficiency
improvement in aviation

- Reduced carbon emissions
- Enhanced vehicle stability

PASSIVE CONTROL = drag reduction without active actuators or extra energy input

Our Three Designs

1) Shark Skin

2) Albatross

3) Boxfish

Shark Skin

Introduction to Shark Skin

Fast-swimming sharks such as the Shortfin Mako can reach speeds up to 45 mph.

- Skin is covered in microscopic tooth-like structures — not smooth rounded scales.
- Hydrodynamic / aerodynamic efficiency: reduces fluid drag during high-speed locomotion.
- Antifouling: helps prevent barnacles, algae and parasites from attaching.



FAST SWIMMER • LOW RESISTANCE

Structure of Shark Skin

1

Anatomy

Millions of microscopic placoid scales called dermal denticles.

2

Composition

Enameloid shell around dentine with a central pulp cavity, anchored in the dermis.

3

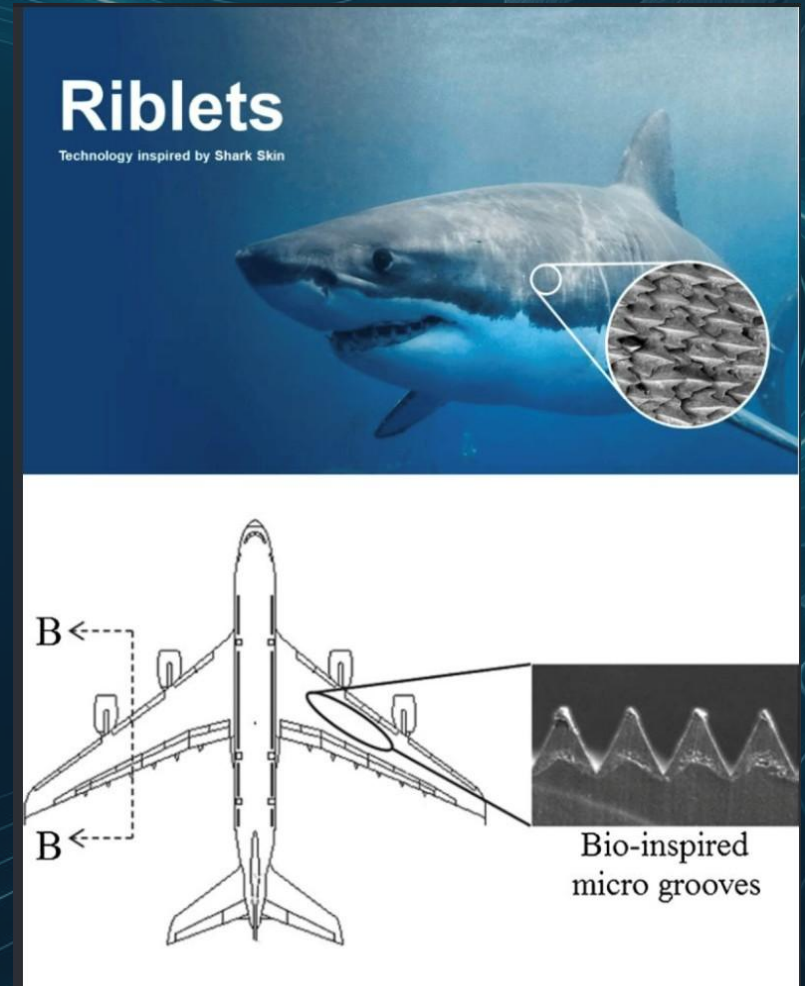
Orientation

Denticles point backward, parallel to the direction of fluid / air flow.

4

Flexibility

Scales can passively erect up to 45° , interacting with turbulent flow to limit separation.



What are Riblets?

Microscopic longitudinal grooves or ridges engineered to mimic the physical profile of dermal denticles.

HEIGHT (h)

SPACING (s)

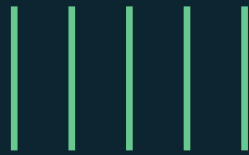
TENS OF μm



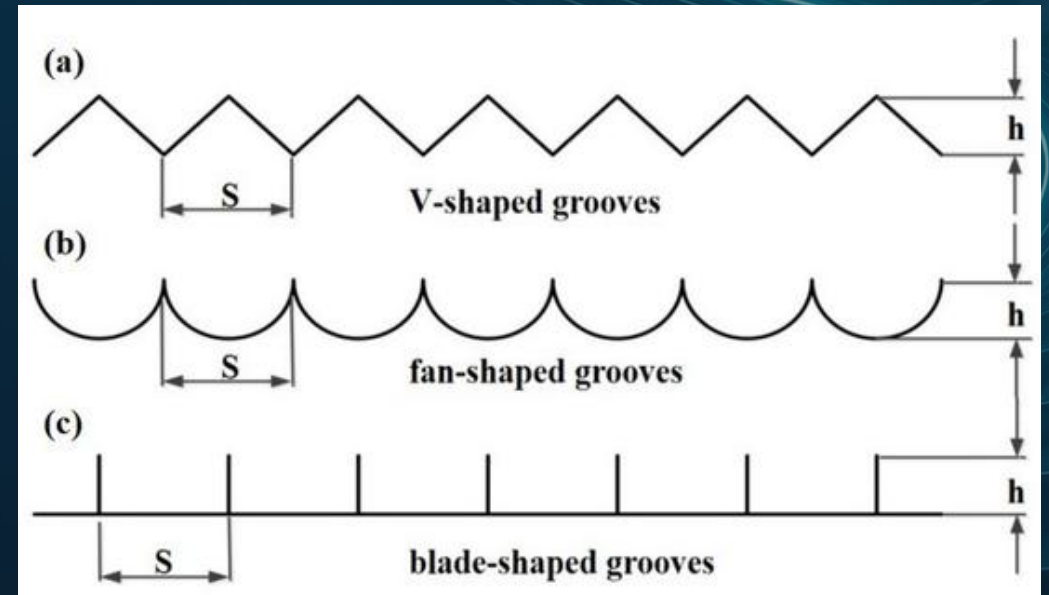
V-shaped



U-shaped



Blade-shaped

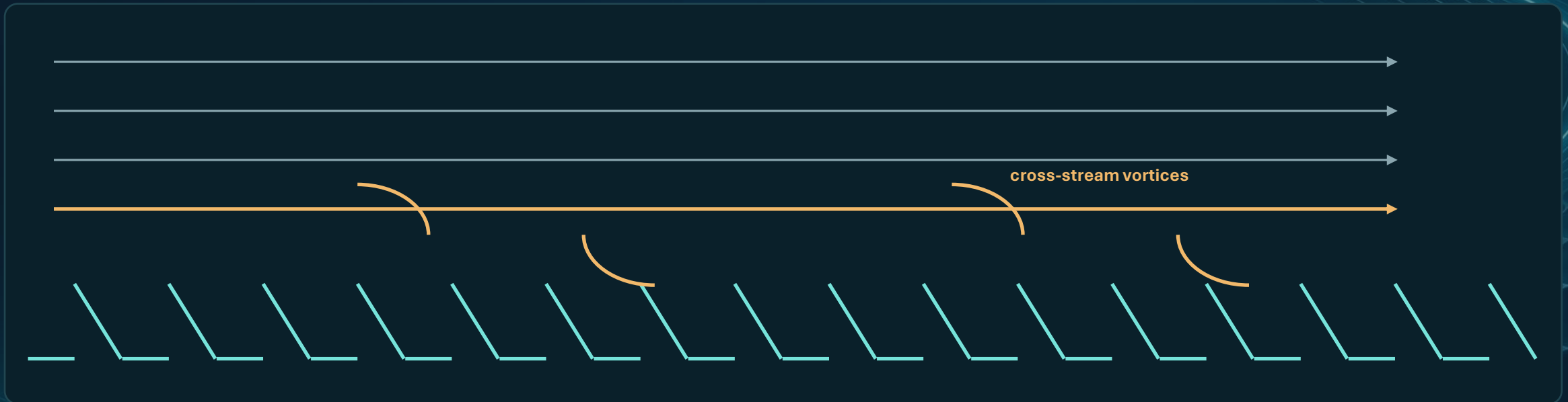


SCALE INVARIANCE

Principles developed in water translate to airflow over aircraft wings and hulls.

How Riblets Affect Airflow

Riblets keep energetic cross-stream vortices away from the valley floor — reducing wall interaction.



LAMINAR → TURBULENT

LOW VELOCITY IN VALLEYS

VORTICES PINNED ABOVE GROOVES

LOWER WALL SHEAR STRESS

Drag Reduction Principle

Skin-friction drag can account for up to 50% of total drag on a commercial aircraft.

5–8%

skin-friction drag reduction

1 Vortex lifting

High-speed fluid is kept away from the valley floor.

2 Cross-stream suppression

Riblets impede spanwise velocity fluctuations (v' , w').

3 Shear-stress mitigation

Wall shear stress τ_w is minimized across much of the surface.

$$\tau_w = \mu(\partial u / \partial y)_{y=0}$$

Smooth Surface vs. Shark-Skin Surface

FEATURE

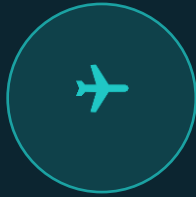
SMOOTH FLAT SURFACE

SHARK-SKIN / RIBLET SURFACE

Vortex surface contact	Sweeps directly against full surface area	Lifted; rests only on riblet tips
Boundary-layer turbulent energy	High turbulent dissipation near wall	Damped cross-stream velocity fluctuations
Wall shear stress (τ_w)	Uncontrolled, maximum friction	Significantly reduced across valleys
Net drag efficiency	Standard baseline	5%–8% drag reduction

Less surface contact → less near-wall energy loss → lower skin-friction drag

Applications in Aircraft and Vehicles



AVIATION

AeroSHARK micro-structured riblet film on Boeing 777F fuselages and engine nacelles.

~1.1% lower fuel burn and CO₂ emissions.



AUTOMOTIVE + MARINE

High-performance racing vehicles and hull coatings for high-speed sea vessels.



WIND TURBINES

Riblet films on turbine blades reduce aerodynamic drag near blade tips and can boost energy output.

Operational deployment: Lufthansa Cargo & SWISS

Research Paper Findings

Walsh et al. • NASA Langley

Wind-tunnel testing demonstrated micro-riblet surfaces can reduce turbulent skin-friction drag by up to

8%
drag reduction

Optimal riblet spacing

Direct Numerical Simulations confirm optimal riblet spacing in viscous wall units is

15–20
viscous wall units

Advantages & Limitations

ADVANTAGES

- Passive control — zero extra energy input
- Scalable application via adhesive films
- Proven fuel-efficiency gains

LIMITATIONS

- Fouling & damage: dirt, dust, insect residue and corrosion degrade micro-groove geometry
- Maintenance & costs: high initial coating cost and complex installation on curved geometries

Future Scope

The next step is not just copying shark skin — it is making the surface adaptive.



Self-cleaning

Superhydrophobic
riblets



Adaptive

Flexible smart
materials



Dynamic pitch

Riblet spacing
changes with
airspeed

Shark-Skin Drag Reduction — Video



Albatross

The Challenge: Efficient Long-Range Flight

Modern UAVs and long-range aircraft consume large amounts of fuel or battery power to stay airborne for long periods.

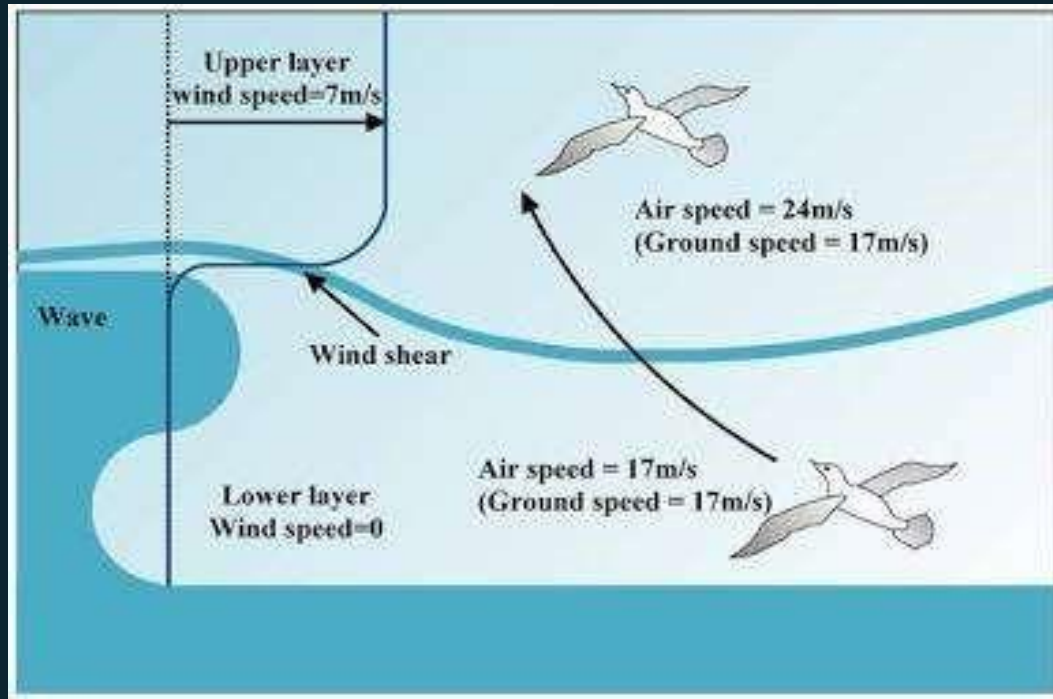
- Engineers struggle to design aircraft that can travel thousands of miles without stopping—especially in high-wind marine environments.
- Conventional wings are rigid and must be heavily reinforced for severe turbulence, adding significant “dead weight”.
- The albatross offers a biological solution: it uses wind energy and flexible outer wings to reduce loads while sustaining flight.



NATURE'S IDEA

Flex with the gust — don't fight it.

Dynamic Soaring: Harvesting Energy from Wind



Dynamic soaring repeatedly crosses air masses of different speeds, allowing the bird to extract energy from the ocean wind gradient.

99%

of flight time can be spent gliding without flapping.

How it works

- The wandering albatross glides for almost its entire flight.
- It continuously exchanges energy with the wind gradient above the ocean.
- This enables sustained flight with minimal wing flapping.

Special Features of Albatross Wings

LONG • NARROW • POINTED

01

Shoulder-lock mechanism

A unique tendon structure locks the wings fully extended, allowing them to stay outstretched without using muscle power.

02

Narrow, pointed shape

The shape reduces air turbulence at the wing tips, minimizing drag.

03

Rigid wing structure

High structural stiffness prevents the wing from twisting under intense wind loads.

From Albatross to Aircraft: AlbatrossOne



Aerospace engineers mimic the albatross to create high-efficiency drones and gliders.

1

ALBATROSS

Outer wing can flex when hit by gusts.

2

HINGED WINGTIP

Freely hinged tips react to turbulence.

3

AIRCRAFT BENEFIT

Structural load on the main wing is reduced.

Aeroelastic Wing Morphing & Hinged Wingtips

STANDARD CRUISE



Wingtips remain locked to maintain high-aspect-ratio efficiency.

STRONG TURBULENCE



The tips unlatch and flex with gusts, isolating the main wing from bending loads.

Result: lighter wing boxes and fuselages can be designed without sacrificing structural integrity.

Why This Design Matters

01

MASSIVE WEIGHT REDUCTION

Flexible tips reduce peak structural stress at the wing root, allowing lighter spars and fuselages.



02

FUEL EFFICIENCY

Lighter aircraft burn less fuel and emit fewer carbon emissions while maintaining long-range capability.



03

LONGER WINGSPANS

Hinged wingtips can fold vertically up to 90° while taxiing, helping solve airport gate limits while keeping long, efficient wings in flight.



Pros & Cons of Albatross-Inspired Designs

PROS

- **Infinite Range**
UAVs can harvest wind energy to stay airborne for weeks without refueling.
- **High Structural Efficiency**
Locking mechanisms reduce the weight of internal actuators and motors.
- **Reduced Carbon Footprint**
High lift-to-drag ratios drastically cut fuel consumption for commercial concepts.

CONS

- **Wind Dependency**
Requires strong, consistent wind gradients; performance falls in calm air.
- **Poor Maneuverability**
Long, high-aspect-ratio wings make quick turns and tight navigation difficult.
- **Large Storage Footprint**
Massive wingspans create challenges for hangar storage and runway spacing.

See the Albatross-Inspired Concept in Action



[Airbus Tests Bird-Like Flapping Wings For Future Aircraft - YouTube](#)

Boxfish

Introduction to the Boxfish

YELLOW BOXFISH • OSTRACION CUBICUS



Rigid shell

**Compact
form**

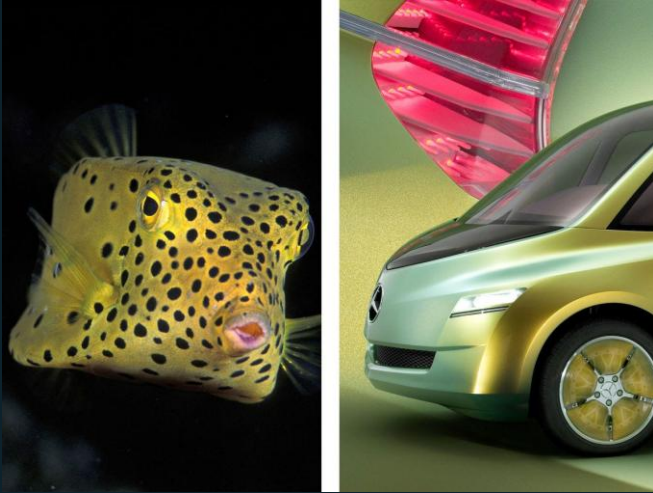
**Agile
swimmer**

A box-shaped fish that challenged aerodynamic expectations.

- Reef-dwelling fish found in the Atlantic, Pacific and Indian Oceans.
- Its squarish body is enclosed in a rigid, lightweight bony shell made from fused hexagonal plates.
- The honeycomb-like shell provides structural strength while keeping a relatively large internal volume.
- Early studies suggested surprisingly low hydrodynamic drag and unusual flow behaviour.
- Despite its box-like appearance, it remains highly manoeuvrable.

Shape & Aerodynamic Characteristics

WHY A BOXFISH CAN STILL MOVE EFFICIENTLY



What stands out

- Short, compact and rigid
- Relatively wide frontal area
- Sharp-ish body edges
- Longitudinal keels / contours

RIGID BONY CARAPACE

Maintains body shape rather than deforming significantly.

COMPACT GEOMETRY

Large internal volume despite a relatively short body.

FLOW SEPARATION

Body geometry affects where fluid detaches and how large the wake becomes.

VORTEX FORMATION

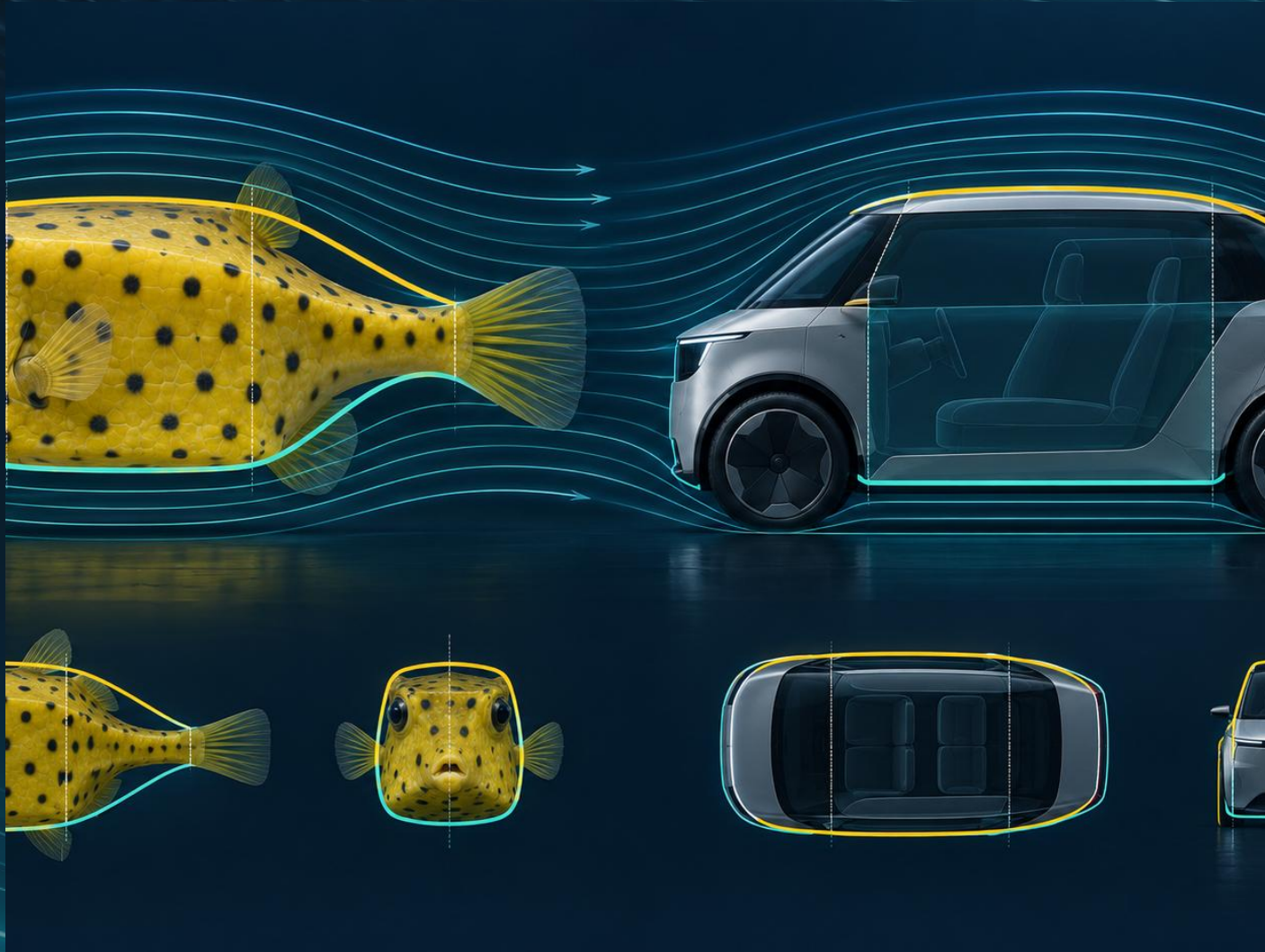
Ridges and keels can generate rotating flow structures that influence forces.

Scientific update

Early work proposed self-correcting stability. Later research found boxfish can also generate destabilizing yaw and pitch moments—possibly supporting manoeuvrability rather than universal passive stability.

How Boxfish Shape Inspired Vehicle Design

FROM BIOLOGICAL FORM → AUTOMOTIVE AERODYNAMICS



What engineers wanted

- Low drag / favourable flow behaviour
- Large internal volume relative to body length
- Rigid, lightweight structure
- Compact form
- Potential for efficient airflow management

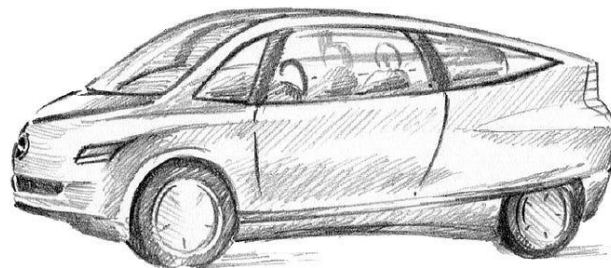
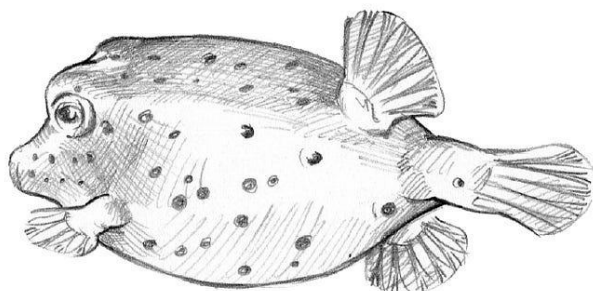
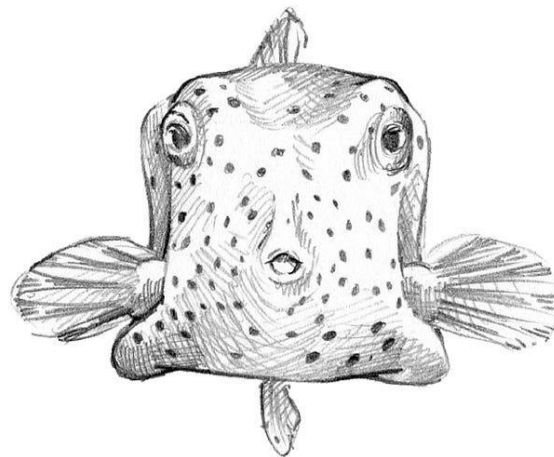
First transfer to a car

A 1:4-scale model borrowed a boxfish-like roof, side contours, side skirts, a descending rear, wedge-shaped rear section and scalloped side surfaces.

Mercedes-Benz Bionic Concept

A 2005 FOUR-SEAT BIOMIMICRY SHOWCASE

BOX FISH CAR



Source: Designing Regenerative Cultures, 2016 - www.danielchristianwahl.com; Graphics: www.flaviagargiulo.com

Concept highlights

2005 unveiled

0.19 reported drag coefficient

~20% lower fuel consumption

up to 80% lower NOx emissions

Airflow: Normal Car vs Boxfish-Inspired Car

FLOW SEPARATION • WAKE • PRESSURE DRAG

Conventional shape



Boxfish-inspired



Abrupt separation → larger turbulent wake → stronger low-pressure pull

Better flow management → controlled separation & wake → lower pressure-drag contribution

Drag Coefficient & Efficiency

THE LOWER THE C_d , THE LOWER THE AERODYNAMIC DRAG

Reported drag-coefficient values



What changes drag?

- Vehicle speed
- Frontal area
- Drag coefficient

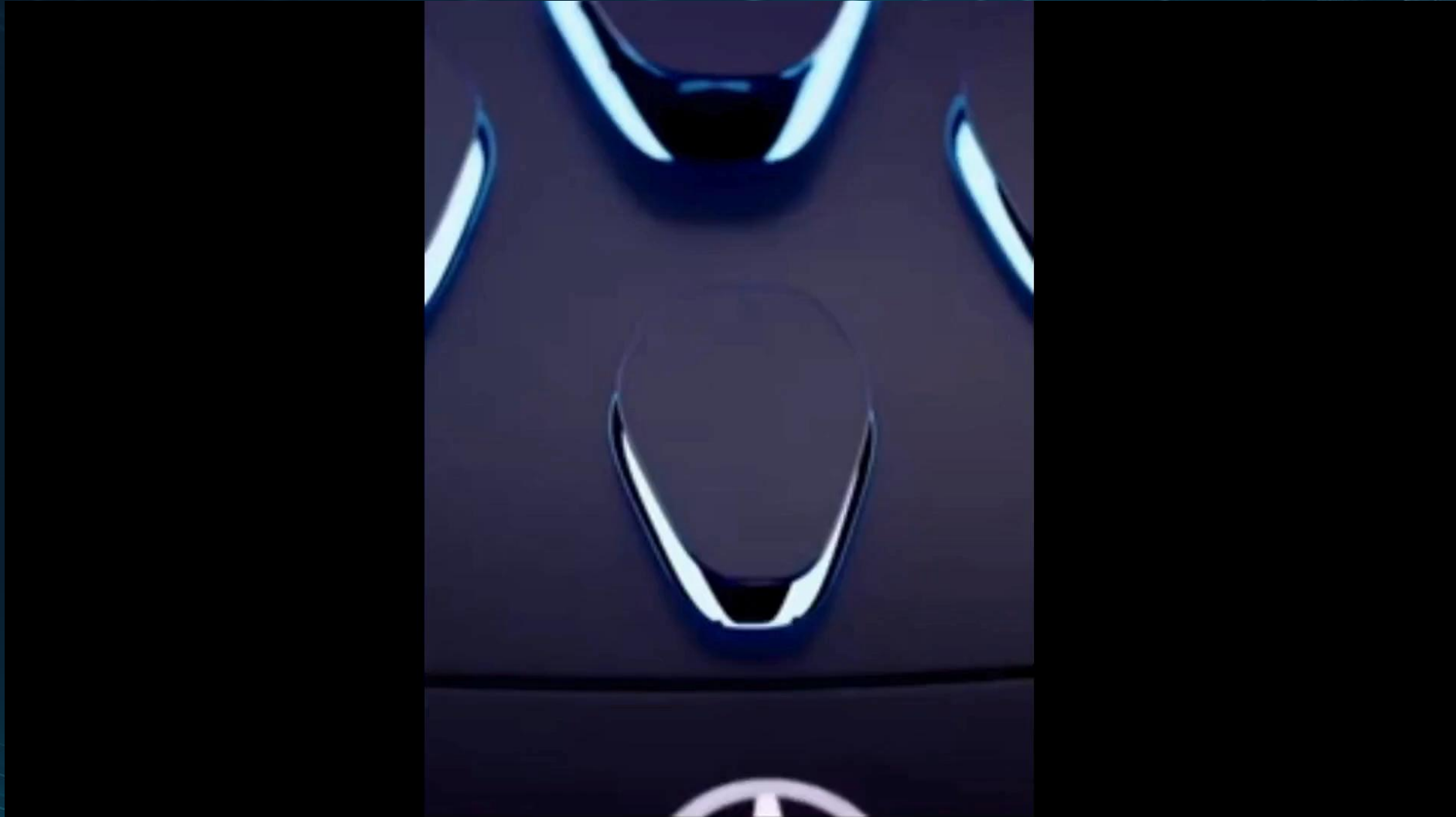
Why the full car is higher

A real vehicle must include wheels, windows, doors, lights, cooling systems, passenger space, structural components and safety requirements.

Important: the boxfish does not have one universal C_d . Results vary with model geometry and test method.

Boxfish Biomimicry in Motion

VIDEO • WATCH THE IDEA COME TO LIFE

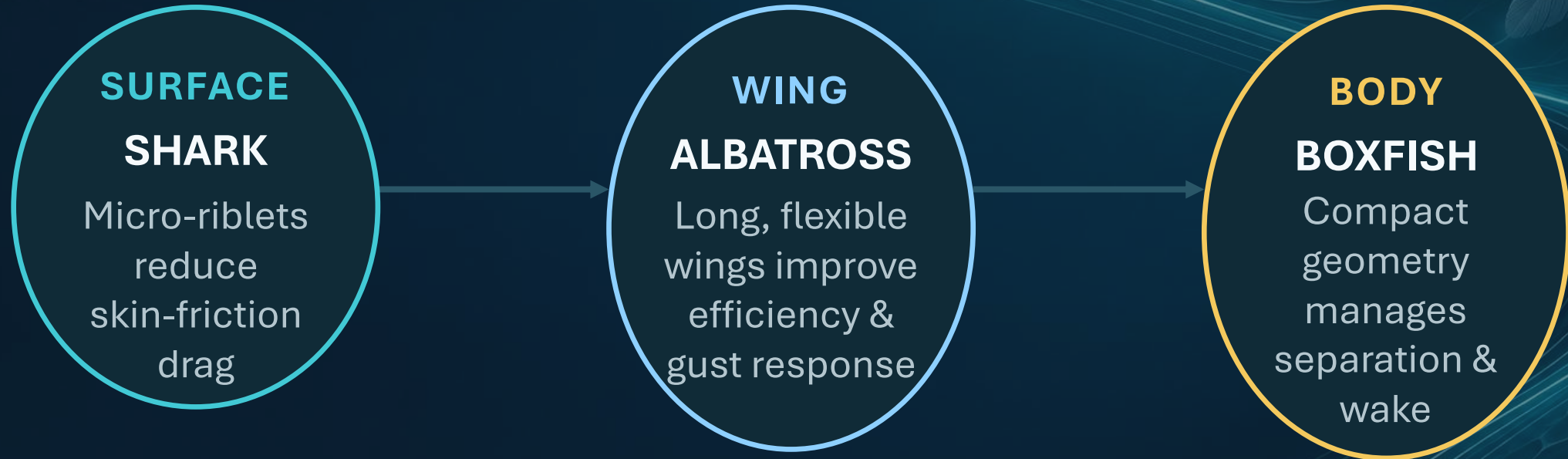


[This Fish inspired a Sports car design from Mercedes - YouTube](#)

Conclusion

Nature solves aerodynamics at three different scales

The strongest lesson is not to copy an animal exactly — it is to translate the useful physical principle into engineering.



KEY TAKEAWAY

Shark skin is the most deployment-ready today; albatross-inspired wings offer major long-range aerospace potential; boxfish shows how unusual body geometry can inspire efficient vehicle forms — but its stability claims must be treated carefully.

Together, they show how biomimicry can reduce drag, lower energy consumption and guide lighter, smarter transportation systems.

THANK YOU

REFERENCES

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- 2) https://www.youtube.com/watch?v=c_zvlwGqZpM
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- 5) <https://www.youtube.com/watch?v=d1z8YUs2FLc>
- 6) <https://slate.com/technology/2015/03/mercedes-benz-bionic-car-boxfish-stability-and-agility-paradox-finally-solved.html>
- 7) <https://www.youtube.com/watch?v=xOQrVQfqE08>